<u>ChatGPT — Article Writing</u>

Artificial Intelligence for Smart Cities

Introduction to Smart Cities

A smart city is an urban area that uses digital technologies to improve the quality of life of its citizens. As the world's population continues to grow, cities face major challenges such as traffic congestion, pollution, waste generation, water shortages, increasing energy demand, and public safety concerns. Traditional methods are often not enough to solve these complex problems. Smart cities address these issues by using technologies such as the Internet of Things, Artificial Intelligence (AI), cloud computing, big data, and communication networks.

AI has become one of the most important technologies in building smart cities. AI enables computers to learn from data, recognize patterns, make predictions, support decision-making without constant human intervention. By combining AI with sensors and connected devices, city authorities can monitor urban activities in real time and provide better public services. As a result, AI is helping cities become more efficient, sustainable, and citizen-friendly.

Role of Artificial Intelligence in Smart Cities

Artificial Intelligence plays a central role in transforming traditional cities into smart cities. Every day, millions of sensors, cameras, GPS devices, and smart meters generate huge amounts of data. AI analyzes this data quickly and provides useful insights that help governments make informed decisions.
Machine learning algorithms can predict traffic conditions, estimate energy demand, identify water leaks, detect security threats, and improve healthcare services. AI also automates repetitive tasks, reducing human effort and increasing operational efficiency. Instead of reacting after a problem occurs, AI allows city administrators to predict problems in advance and take preventive actions.
For example, AI systems can detect unusual traffic patterns before congestion becomes severe or identify equipment failures before they cause service interruptions. This predictive capability makes city management more efficient and reliable.

AI Applications in Smart Cities

1. AI in Traffic Management

Traffic congestion is one of the biggest problems in urban areas. AI helps manage traffic by analyzing live data from CCTV cameras, GPS devices, road sensors, and traffic signals.

AI-powered traffic systems can:

- Optimize traffic signal timings based on vehicle density.
- Suggest the fastest routes using real-time traffic information.
- Detect road accidents and notify emergency services immediately.
- Improve public transportation schedules.

Example: Bengaluru and Delhi use AI-enabled traffic monitoring systems to reduce congestion at busy intersections. These systems analyze vehicle movement and automatically adjust traffic signals to improve traffic flow.

2. AI in Smart Energy Management

Energy consumption continues to increase with urban development. AI helps cities use electricity more efficiently by analyzing consumption patterns and predicting future demand.

Applications include:

- Smart grids that balance electricity supply and demand.
- AI-controlled street lighting that automatically adjusts brightness depending on traffic and weather conditions.
- Predictive maintenance of power equipment.
- Integration of renewable energy sources like solar and wind.

Example: Several smart city projects, including Pune Smart City, have introduced intelligent LED street lighting systems that reduce electricity consumption and maintenance costs through automated monitoring.

3. AI in Waste Management

Efficient waste collection is essential for maintaining clean cities. Traditional waste collection follows fixed schedules, which often results in unnecessary trips or overflowing bins.

AI improves waste management by:

- Monitoring garbage bin levels using smart sensors.
- Planning optimized collection routes.
- Predicting waste generation trends.
- Supporting automated waste segregation and recycling.

Example: Indore, one of India's cleanest cities, has adopted smart waste management systems with GPS tracking and digital monitoring to improve waste collection efficiency and cleanliness.

4. AI in Water Supply Management

Water is a valuable resource, and many cities face water shortages due to increasing population and leakage in distribution systems. AI helps improve water management by monitoring pipelines, reservoirs, and consumption patterns.

AI applications include:

- Detecting pipeline leaks.
- Predicting future water demand.
- Monitoring water quality in real time.
- Reducing water wastage through smart distribution.

Example: Smart water management initiatives under the Smart Cities Mission use IoT sensors and AI-based monitoring to detect leakage and improve water distribution in cities such as Surat.

5. AI in Healthcare

Healthcare is another important area where AI improves smart city services. AI enables faster diagnosis, remote monitoring, and better hospital management.

Applications include:

- AI-assisted medical diagnosis.
- Predicting disease outbreaks using health data.
- Remote patient monitoring through wearable devices.
- Managing hospital resources efficiently.

Example: During the COVID-19 pandemic, several Indian states used AI tools for tracking infection trends, monitoring quarantine cases, and supporting healthcare planning.

6. AI in Public Safety

Ensuring public safety is a major responsibility of city administrations. AI helps improve security by continuously monitoring public spaces and identifying unusual activities.

Applications include:

- Facial recognition systems.
- Intelligent video surveillance.
- Crime pattern analysis.
- Emergency response management.
- Disaster prediction and early warning systems.

Example: Hyderabad's Integrated Command and Control Centre uses AI-enabled surveillance cameras and analytics to improve traffic regulation, crime prevention, and emergency response.

Benefits of AI for Citizens and Government

Artificial Intelligence offers several advantages for both citizens and government authorities.

Benefits for Citizens

- Reduced traffic congestion and shorter travel times.
- Better healthcare services with faster diagnosis.
- Improved water and electricity supply.
- Cleaner environment through efficient waste management.
- Increased public safety using intelligent surveillance.
- Faster emergency response during accidents or disasters.
- Better quality of life through efficient city services.

Benefits for Government

- Better decision-making based on real-time data.
- Reduced operational and maintenance costs.
- Improved resource utilization.
- Predictive maintenance of public infrastructure.
- Better planning for future urban development.
- Increased transparency and efficient public service delivery.

Challenges of AI in Smart Cities

Although AI provides many advantages, several challenges must be addressed before large-scale implementation.

- **Privacy Issues:** Smart cities collect massive amounts of personal data through cameras, mobile devices, and sensors. Protecting citizens' privacy and preventing misuse of personal information is a major concern.
- **Cybersecurity Risks:** Connected systems may become targets for cyberattacks. A successful attack on transportation, electricity, or healthcare infrastructure could seriously affect public services. Strong cybersecurity measures are essential.
- **High Implementation Cost:** Building AI-enabled infrastructure requires significant investment in sensors, communication networks, cloud platforms, data centers, and skilled professionals. Many developing cities face financial limitations.
- **Ethical Issues:** AI systems may sometimes produce biased decisions due to biased training data. Excessive surveillance can also create concerns about individual freedom and human rights. Governments must ensure that AI systems remain transparent, fair, and accountable.
- **Skill Gap:** Implementing AI requires trained professionals in data science, machine learning, cybersecurity, and cloud computing. Developing these skills remains an important challenge for many cities.

Future Scope of AI in Smart Cities

The future of Artificial Intelligence in smart cities is highly promising. As AI technologies continue to improve, cities will become increasingly automated, connected, and sustainable.

Future developments may include:

- Fully autonomous public transportation systems.
- AI-powered digital assistants for government services.
- Intelligent disaster prediction and emergency management.
- Smart environmental monitoring to reduce pollution.
- AI-driven urban planning using predictive analytics.
- Integration of 5G, IoT, cloud computing, edge computing for faster decision-making.
- Personalized citizen services based on real-time needs.

Conclusion

Artificial Intelligence is becoming one of the key technologies for building modern smart cities. By analyzing large amounts of real-time data, AI helps improve transportation, energy management, healthcare, waste collection, water distribution, and public safety. These improvements lead to better public services, reduced operational costs, and an enhanced quality of life for citizens.

Despite challenges such as privacy concerns, cybersecurity threats, implementation costs, and ethical issues, the benefits of AI are significant. With proper policies, strong data protection, skilled professionals, and responsible implementation, AI can create safer, cleaner, and more sustainable cities. As India continues its digital transformation through initiatives like the Smart Cities Mission, Artificial Intelligence will remain a driving force in shaping the cities of the future.

Gemini — Research on AI in Traffic Management

Literature Review: AI in Traffic Management for Smart Cities

1. Current Urban Traffic Challenges

Modern metropolitan centers face unprecedented mobility strain due to rapid urbanization, population density spikes, and aging infrastructure. Key issues include:

- **Severe Gridlock & Delay:** Fixed-timer traffic lights fail to respond to real-time traffic volume, creating persistent bottlenecks during peak hours.
- **Increased Carbon Emissions:** Stop-and-go driving patterns significantly elevate greenhouse gas (CO_2) emissions and localized air pollution.
- **Emergency Delays:** Traditional traffic systems cannot dynamically prioritize emergency vehicles, leading to life-threatening dispatch delays.
- **Economic Losses:** Operational inefficiency costs billions annually in wasted fuel, lost productivity, and delayed logistics.

2. Core Technological Enablers of AI in Mobility

AI transforms raw urban data into automated decision-making using four foundational technology pillars:

- **Computer Vision (CV):** Video surveillance cameras powered by deep learning models analyze live video feeds to detect vehicle density, classify vehicle types, identify road hazards, and spot traffic violations without manual human monitoring.
- **IoT Sensors & Edge Computing:** Distributed inductive loops, radar sensors, and connected vehicle (V2X) nodes continuously gather speed and volume data. Edge-AI devices process this telemetry directly at the intersection node to minimize latency.
- **Reinforcement Learning (RL):** RL agents (such as Deep Q-Networks) model intersections as dynamic state environments. The system continuously experiments with signal phase durations to maximize vehicle throughput and minimize delay penalties.
- **Predictive Analytics:** Machine learning models process temporal-spatial traffic data combined with external variables to forecast gridlock up to hours before it manifests.

3. Practical AI Applications in Smart Traffic Systems

- **Smart Traffic Signals:** Unlike legacy pre-timed systems, AI-driven traffic signals adjust green light phases dynamically. By processing vehicle queues in real time across connected intersections, signal controllers optimize green-wave corridors, dramatically lowering total idle time.
- **AI-Based Accident Detection & Violation Monitoring:** Computer vision models analyze traffic feeds for anomalous motion patterns—such as sudden deceleration, lane deviation, or vehicle stoppages. Upon detection, automated alerts are instantly dispatched to emergency services while log entries document infractions like red-light violations and speeding.
- **Emergency Vehicle Priority Systems (EVPS):** AI integrates GPS telemetry from first responders with smart signal corridors. When an emergency vehicle approaches an intersection, the system preempts normal light cycles, clearing downstream queues and securing an uninterrupted green corridor.
- **Traffic Prediction & Dynamic Routing:** Using Recurrent Neural Networks (RNNs) and Graph Neural Networks (GNNs), systems analyze urban transit flow to predict arterial congestion. Route guidance systems then proactively redistribute vehicle navigation suggestions to balance network load.

4. Real-World Case Studies (2023–2026)

Case Study 1: Singapore — AI-Powered Digital Twin & ITS
Singapore has deployed dynamic traffic modeling alongside its Virtual Singapore digital twin framework. Integrating vehicle-to-everything (V2X) telemetry with generative urban simulations, the city dynamically recalculates traffic light timings and manages demand pricing in real time, yielding smoother transit flows and reduced intersection congestion across high-density corridors.
Case Study 2: Los Angeles, USA — Smart Emergency Priority & Traffic Orchestration
Under its Smart Mobility initiative, Los Angeles integrated AI-driven traffic signals with emergency dispatch systems. By continuously monitoring traffic camera feeds and tracking first-responder GPS coordinates, the system automatically adjusts signal phasing to establish cleared routes for emergency vehicles, cutting first-responder response times by up to 30%.
Case Study 3: Barcelona, Spain — Adaptive Signal Control & Emission Reduction
Barcelona implemented machine learning-driven signal management systems across key urban arteries. By processing multi-modal traffic data from IoT sensors, the system continuously tailors signal timing to priority corridors. The deployment achieved a 20% reduction in average travel delay times alongside a 15% drop in vehicular emissions.

5. Comparative Analysis: Traditional vs. AI-Based Traffic Management

Dimension	Traditional Traffic Management	AI-Based Traffic Management
Signal Phasing	Fixed time schedules or rule-based timers	Dynamic adaptation based on real-time vehicle queues
Data Source	Historical counts and manual surveys	Live video feeds, IoT sensors, GPS, and V2X telemetry
Incident Response	Manual dispatch based on driver reports	Automated real-time anomaly detection and instant alerts
Emergency Priority	Non-existent or manual siren detection	Automated smart corridor preemptive green lights
Network Coordination	Isolated intersections with rigid green waves	City-wide multi-agent RL optimization across entire grids
Environmental Impact	High idle times and elevated emissions	Optimized speed profiles reducing fuel burn and CO_2

6. References (IEEE Format)

1. V. N. Skoropad, "Dynamic traffic flow optimization using reinforcement learning and predictive analytics: A sustainable approach to improving urban mobility in the city of Belgrade," *Sustainability*, vol. 17, no. 8, Art. no. 3383, 2025.
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3. D. Fischer, "Deep learning-based predictive traffic signal optimization for smart cities: A comparative analysis of centralized and edge-AI urban transportation architecture," *Journal of Emerging Technology and Engineering*, vol. 12, no. 2, pp. 45–58, 2025.
4. A. L. Maureal, "A lightweight edge AI framework for adaptive traffic signal control in mid-sized Philippine cities," *Sustainability*, vol. 18, no. 3, Art. no. 1147, 2026.
5. R. Ithrees, "AI-based intelligent traffic management system," *International Journal of Scientific Research and Technology*, vol. 3, no. 4, pp. 846–858, 2026.

Perplexity — Compare Normal Cities and Smart Cities

Normal Cities vs AI-Enabled Smart Cities

Introduction

Cities today are changing from traditional systems to technology-driven urban ecosystems. Normal cities depend mostly on manual management and conventional infrastructure, while AI-enabled smart cities use data, sensors, and automation to improve services and planning. This comparison shows how both city models differ across key areas of urban life.

Comparison Table

Parameter	Normal Cities	AI-Enabled Smart Cities
Infrastructure	Conventional infrastructure with limited digital integration.	Connected infrastructure using sensors, IoT, and intelligent systems.
Traffic management	Traffic is managed mostly through fixed signals and manual control.	Real-time traffic monitoring and predictive control improve flow and reduce congestion.
Energy consumption	Energy use is less optimized and often wasteful.	Smart grids and analytics help reduce energy loss and improve efficiency.
Waste management	Waste collection is usually schedule-based and less data-driven.	Smart waste systems track bins, routes, and collection needs more efficiently.
Water management	Water supply and leakage control are often less precise.	Smart meters and monitoring systems improve water use and reduce loss.
Public safety	Safety systems rely mainly on traditional policing and manual response.	AI-based alerts, surveillance, and analytics improve response and prevention.
Citizen services	Services often require in-person visits and more paperwork.	Digital platforms make services faster, easier, and more accessible.
Environmental impact	Usually higher pollution and resource waste.	Better resource control can reduce emissions and environmental damage.
Cost of operation	Lower initial technology cost, but higher inefficiency over time.	Higher setup cost, but better long-term efficiency and savings.
Data usage	Limited or fragmented data use.	Heavy use of real-time data for planning and decision-making.

Summary

Normal cities and AI-enabled smart cities differ mainly in the way they manage urban systems. Normal cities usually depend on traditional methods, manual monitoring, and slower decision-making, which can lead to congestion, higher waste, and less efficient service delivery. In contrast, smart cities use digital technology, sensors, and artificial intelligence to collect data and improve transportation, energy use, waste handling, water management, and public safety. These technologies help city administrators respond faster and plan more effectively.

Smart cities also support better citizen services through mobile apps, online platforms, and connected systems, making daily life more convenient. They can reduce environmental impact by cutting energy waste, improving air quality, and using resources more carefully. However, smart cities need large investments, strong digital infrastructure, and proper management of data and privacy. Therefore, while normal cities are simpler and cheaper to run initially, AI-enabled smart cities offer a more efficient, sustainable, and future-ready model of urban development.

Advantages of Smart Cities

1. Better traffic and mobility management through real-time data and AI.
2. More efficient use of energy, water, and waste systems.
3. Faster and more convenient citizen services through digital platforms.

Limitations of Smart Cities

1. High setup and maintenance cost for technology infrastructure.
2. Dependence on data systems, which creates privacy and security concerns.
3. Benefits may be uneven if access and inclusion are not properly planned.

Conclusion

AI-enabled smart cities represent the next stage of urban development by making cities more efficient, responsive, and sustainable. Compared to normal cities, they offer better management of traffic, services, and resources, but they also require careful planning, investment, and responsible use of technology. In the future, the best cities will likely combine smart technology with human-centered governance to improve quality of life for everyone.

Web sources

- IMD Smart City Index 2026: https://www.imd.org/smart-city-observatory/home/
- Smart City vs. Normal City: Understanding the Differences: https://minnovation.com.au/smart-cities-2/smart-city-vs-normal-city-understanding-the-differences/
- Smart traffic management research: https://pmc.ncbi.nlm.nih.gov/articles/PMC5134531/
- Smart cities and environmental sustainability: https://journals.rta.lv/index.php/ETR/article/view/7948
- Smart city environmental and energy impact article: https://www.sciencedirect.com/science/article/pii/S0306261922003087
- IoT-based smart waste management in cities: https://www.verizon.com/business/resources/articles/s/iot-based-smart-waste-management-in-cities/

Claude — Report Analysis & Summary

Policy Analysis: Report on Smart City Mission–India (SESEI, July 2018)

Executive Summary (150 words)

India's Smart Cities Mission, launched in June 2015 under PM Narendra Modi, aims to develop 100 smart cities by 2024 through a competitive, area-based development approach combining retrofitting, redevelopment, greenfield development, and pan-city solutions. Selection concluded in June 2018 via a two-stage challenge process, with all states and union territories (except West Bengal) nominating cities. The Mission is centrally sponsored, with a total estimated outlay of ₹2.05 lakh crore (~€25.6 billion), implemented through city-level Special Purpose Vehicles (SPVs). Despite strong ambitions, implementation lags: only 6.4% of projects were complete by early 2018, utilizing just 1.6% of envisaged investment. Key challenges include financing gaps, weak ULB capacity, absent master plans, and legacy infrastructure. The report highlights IoT/M2M as central "horizontal" enablers of smart energy, transport, waste, and governance solutions, alongside international partnerships (Germany, Japan, France, Spain, Singapore) and standardization efforts by BIS, TSDSI, and SESEI.

Main Objectives of the Report

- Document the origin, structure, and governance framework of India's Smart Cities Mission (SCM)
- Explain the city selection process (Smart City Challenge, two-stage competition)
- Track financing mechanisms and fund disbursement to selected cities
- Assess implementation status and progress as of early-mid 2018
- Identify barriers to effective smart city development
- Map convergence between SCM and complementary schemes (AMRUT, HRIDAY, Swachh Bharat Mission)
- Survey the role of IoT/M2M and ICT as core technological enablers
- Document India's standardization ecosystem (BIS, TSDSI, MeitY, MoC) and SESEI's role in India-EU cooperation on smart city standards

Key AI/Technology Themes Mentioned

(Note: the report focuses on IoT/M2M and ICT broadly rather than AI specifically)

- **IoT (Internet of Things)** – sensor-based data collection across city systems
- **M2M (Machine-to-Machine) communication** – enabling intelligent, real-time decision-making (traffic lights, parking, energy)
- **oneM2M platform** – global standards initiative; C-DOT's indigenous CCSP (C-DOT Common Service Platform) built on this standard
- **Smart Grids & Smart Meters** – for energy monitoring and distribution
- **Smart Data/Big Data Analytics** – predictive analysis of city service data
- **Smart Parking, Smart Waste Management, Smart Lighting** – sensor-driven automation
- **Intelligent Traffic Management Systems** – for congestion and safety
- **Device connectivity & data mobility** – interoperability across administrative/municipal systems raising security and privacy considerations

Important Statistics and Findings

Metric	Value
Total cities targeted	100 (99 selected by report date + 1 pending)
Target completion year	2024
Total Mission budget (5 years)	₹480 billion (~€6 billion) central; ~₹2.05 lakh crore (~€25.6bn) total proposed investment
Central assistance per city	₹100 crore/year (first year: ₹200 crore)
Matched state/ULB funding	Equal (50:50) contribution required
Projects completed (Jan 2018)	189 projects worth ₹2,237 crore (6.4% of total)
Investment utilized	Just 1.6% of ₹1,38,730 crore envisaged
Projects at DPR stage only	1,987 projects (₹1,01,992 crore) — the majority
Cities without master plans	70–80% of Indian cities
Urban population (India, 2018)	30% of 1.35 billion; generates ~2/3 of GDP
Major funding partners	Spain, USA, Germany, Japan, France, Singapore, Sweden

Benefits for Urban Development

- **Structured, competitive selection** encourages cities to build genuine local development strategies
- **Area-based development models** (retrofitting, redevelopment, greenfield) offer flexibility suited to different urban contexts
- **Convergence with AMRUT, HRIDAY, Swachh Bharat** maximizes synergy and avoids duplicated infrastructure spending
- **Citizen participation** in SCP formulation promotes a "bottom-up" planning approach
- **International technology transfer** through partnerships with Germany, Japan, France, Singapore, etc.
- **IoT/M2M-driven services** (smart energy, traffic, waste) can materially improve efficiency and quality of life
- **Standardization efforts** (BIS, TSDSI, oneM2M) aim to prevent vendor lock-in and ensure interoperability
- **Liveability Index** creates a competitive, data-driven benchmarking system across 116 cities

Challenges Identified

1. **Financing gap** – annual requirement (~₹35,000 crore) far exceeds available central/state funds; heavy reliance on PPPs
2. **Financial unsustainability of ULBs** – tariffs don't reflect actual service costs
3. **Absence of master plans** – 70–80% of Indian cities lack basic development plans
4. **Legacy infrastructure retrofitting** – difficult integration of old systems with new smart solutions
5. **Weak technical capacity of ULBs** – limited skilled staffing, poor market-competitive recruitment

6. **Fragmented three-tier governance** – coordination gaps between central, state, and local bodies
7. **Slow regulatory clearances** – lack of time-bound, online clearance processes
8. **Multivendor technology environments** – interoperability challenges across different suppliers
9. **Capacity-building shortfalls** – only ~5% of funds allocated to training/knowledge-building
10. **Unreliable utility services** – inconsistent 24x7 power/water supply undermines "smart" upgrades
11. **Citizen resistance** to user charges for new services
12. **Slow implementation pace** – vast majority of projects stuck at planning/DPR stage rather than execution

Recommendations for Future Improvement

- **Accelerate fund utilization**: streamline disbursement and reduce the DPR-to-execution bottleneck
- **Strengthen ULB financial autonomy**: reform tariff structures to reflect true service costs; expand municipal bond markets
- **Mandate city master plans**: prioritize development plans as a precondition for further funding tranches
- **Invest more in capacity building**: increase allocation beyond ~5% for training and technical skill development
- **Establish sector regulators**: create independent bodies for utility services to balance private participation with fair tariffs
- **Adopt standardized, interoperable ICT architecture**: leverage BIS/TSDSI/oneM2M frameworks to avoid vendor lock-in
- **Improve inter-governmental coordination**: formalize mechanisms between MoUD, states, and ULBs for faster clearances
- **Expand PPP and innovative financing**: scale up Tax Increment Financing, Pooled Finance Development Funds, and leverage NIIF
- **Continue international collaboration**: deepen technology and knowledge partnerships (EU-India via SESEI, Indo-Canadian training initiatives)
- **Monitor via Liveability Index**: use ongoing city rankings to drive competitive, outcome-based improvement

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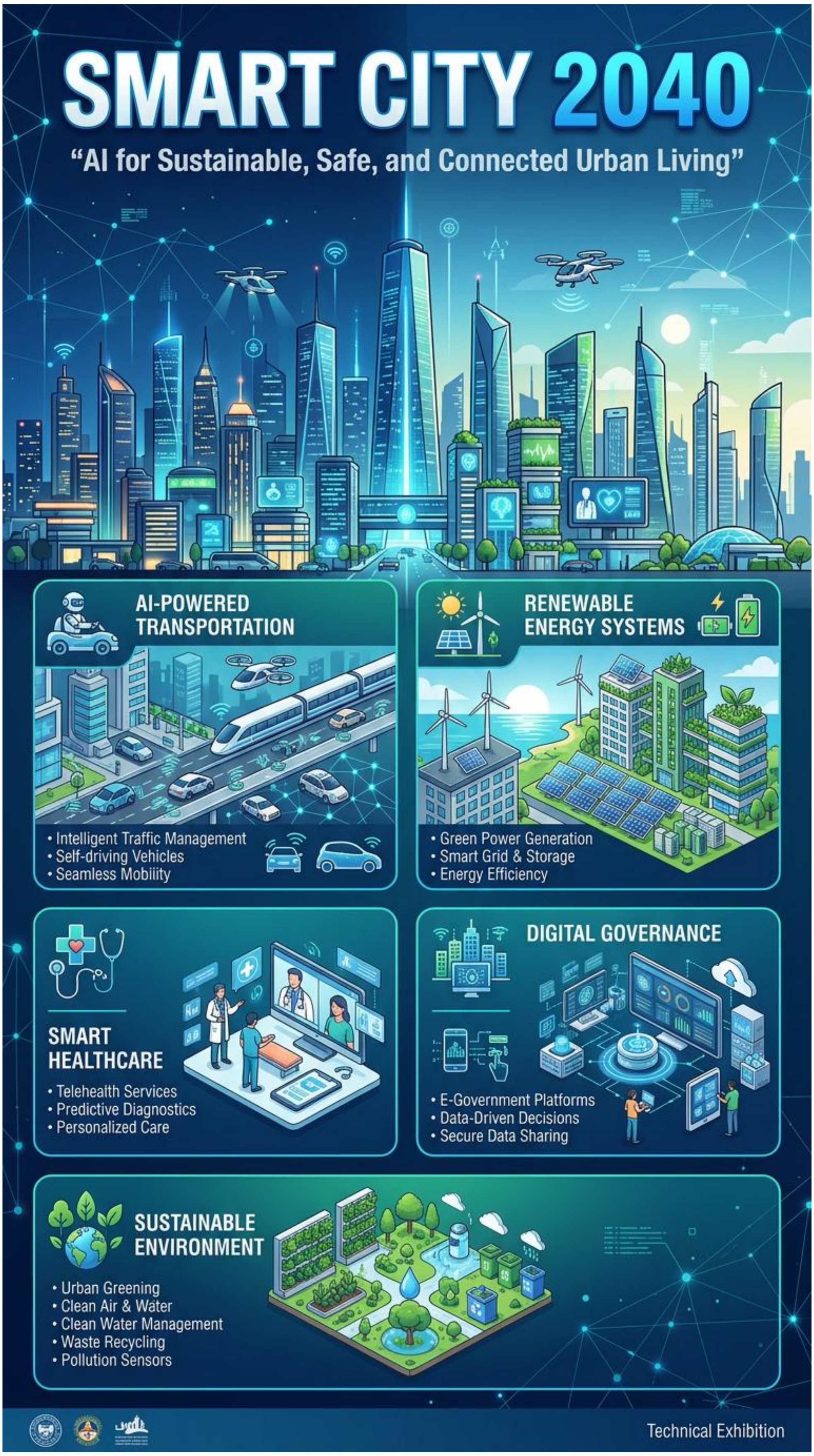
DALL·E — AI-Powered Smart City Image

Adobe Firefly — Poster "Smart City 2040"



Canva AI — Infographic

Microsoft Copilot — Smart Parking System Code

```
# Smart Parking System
# Total parking slots = 20
# Using dictionary to store vehicle number and slot number

# Initialize parking slots (empty dictionary)
parking_slots = {}
TOTAL_SLOTS = 20

# Function to park a vehicle
def park_vehicle(vehicle_number):
    if vehicle_number in parking_slots.values():
        print("Error: Vehicle already parked!")
        return
    if len(parking_slots) >= TOTAL_SLOTS:
        print("Parking Full! No slots available.")
        return
    # Find the first available slot
    for slot in range(1, TOTAL_SLOTS + 1):
        if slot not in parking_slots:
            parking_slots[slot] = vehicle_number
            print(f"Vehicle {vehicle_number} parked at slot {slot}.")
            break

# Function to remove a vehicle
def remove_vehicle(vehicle_number):
    for slot, v_number in list(parking_slots.items()):
        if v_number == vehicle_number:
            del parking_slots[slot]
            print(f"Vehicle {vehicle_number} removed from slot {slot}.")
            return
    print("Error: Vehicle not found!")

# Function to display available slots
def display_available_slots():
    available = [slot for slot in range(1, TOTAL_SLOTS + 1) if slot not in
parking_slots]
    print("Available slots:", available)

# Function to search for a vehicle
def search_vehicle(vehicle_number):
    for slot, v_number in parking_slots.items():
        if v_number == vehicle_number:
            print(f"Vehicle {vehicle_number} is parked at slot {slot}.")
            return
    print("Vehicle not found!")

# Sample usage
```

```python
if __name__ == "__main__":
    # Park some vehicles
    park_vehicle("TN01AB1234")
    park_vehicle("TN02XY5678")
    park_vehicle("TN03ZZ9999")

    # Try duplicate entry
    park_vehicle("TN01AB1234")

    # Display available slots
    display_available_slots()

    # Search for a vehicle
    search_vehicle("TN02XY5678")
    search_vehicle("TN05AA1111")

    # Remove a vehicle
    remove_vehicle("TN02XY5678")

    # Display slots again
    display_available_slots()
```